

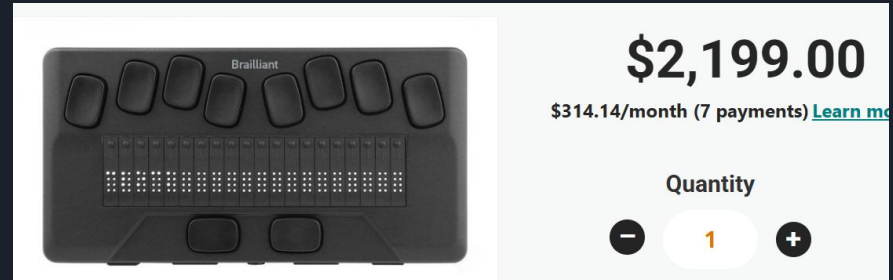


E-Braille

Annie, Fara, Gina, Inaya

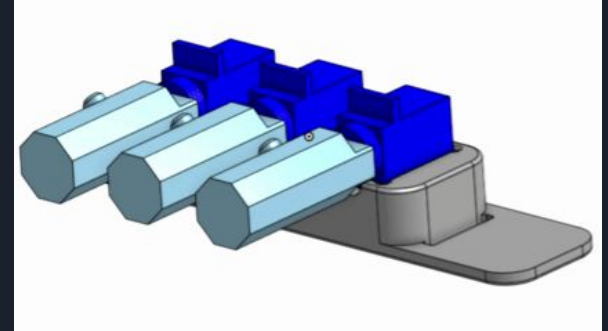
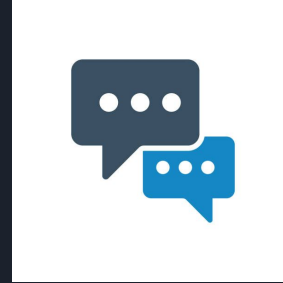
Brainstorming

- We have a hardware inclined skillset
- Blind people do not have much accessible communications methods
- Current solutions:
 - Very expensive (\$2000+)
 - Closed-sourced design



Solution – A Braille Texter!

- A pair of devices that sends messages in Braille and displays messages in Braille!
- Input: Push Buttons → Tactile!
- Output: Braille Rollers driven by servo motors → Also tactile!
- Costs around \$150 in total
 - Significantly less!



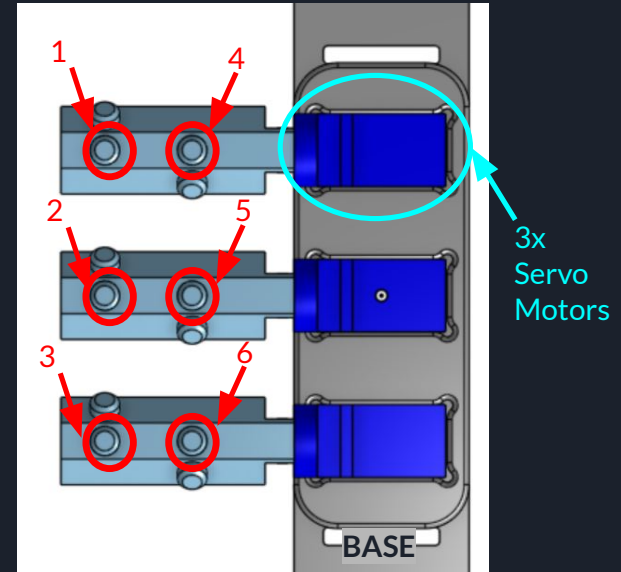
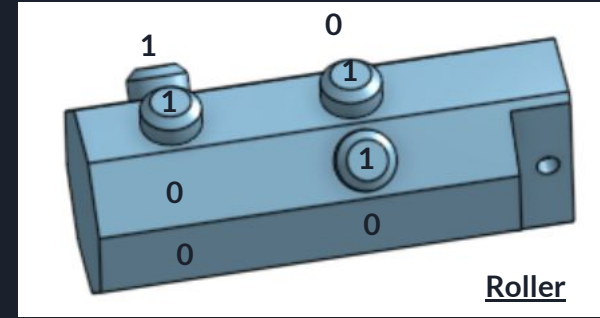
Mechanical

Process

1. Computer Aided Design (Onshape)
2. 3D printed rollers and base

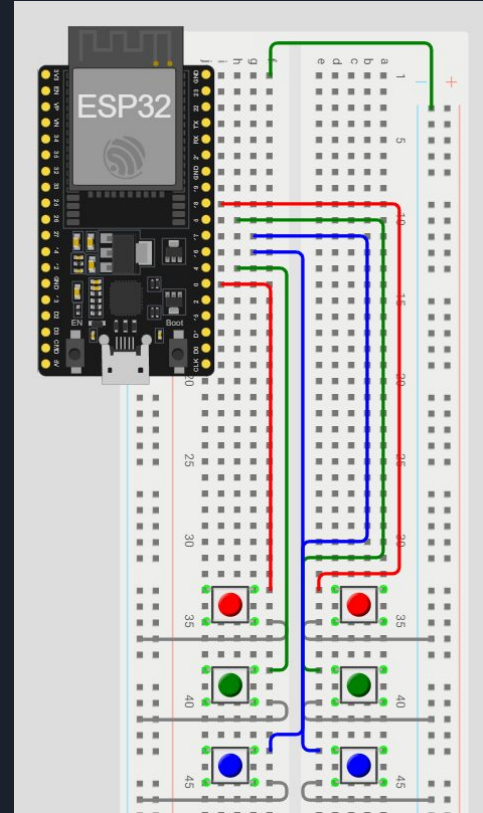
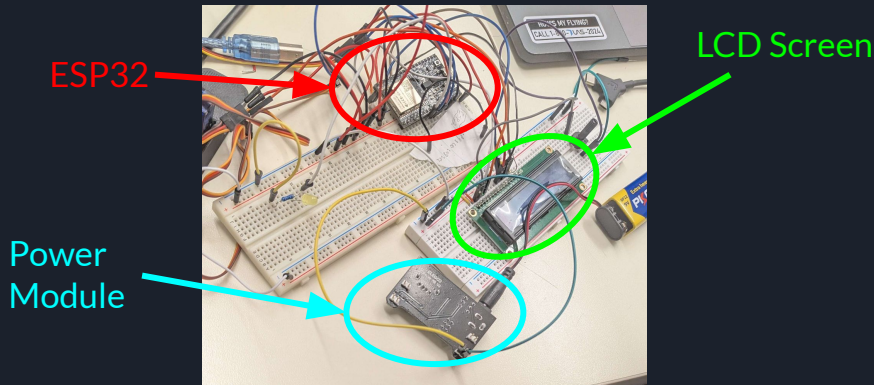
Design

- Each servo motor outputs 4 positions
 - Corresponds to 00, 01, 10, 11 in binary
- 3 rollers → 26 alphabets + punctuation



Electrical

- We used ESP32 modules for wireless communication
- LCD screen for debugging/ secondary display
- LEDs as status lights
- Secondary power source

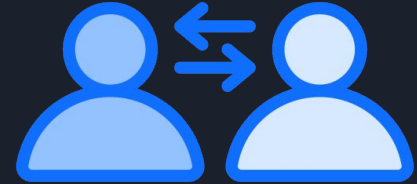




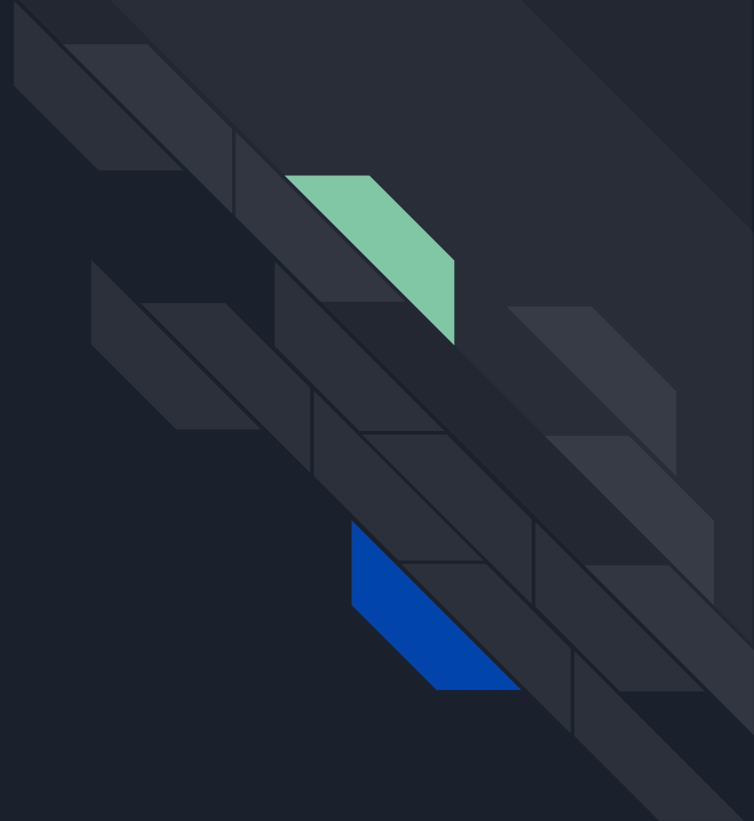
Demo

Further Improvements

- Miniaturize the mechanism
- Allowing both ESP32 modules to send and receive Braille
- Combine multiple Braille units to create a multi-character display



Thank You!





Appendix A – Additional Project Documents

- [Onshape CAD File](#)
- [Wokwi Electrical Schematic](#)
- [GitHub Repository](#)



Appendix B – Cost Approximation

- 66.6g of 3D printing PLA * \$0.04/g = \$2.67
- 2 ESP32 modules = \$60
- 6 servo motors = \$15
- 2 Arduino Sensor Kits = \$70
- 14 push buttons = \$1.12

Total = \$150